

ENGINEERING INSPECTION REPORT

Report ID	ENG-INS-2026-001
Inspection Date	07 September 2026
Inspection Type	Equipment Condition and Surface Inspection
Facility	ABC Industrial Manufacturing Unit
Inspector	Arun Kumar
Location	Coimbatore, Tamil Nadu

Inspection Findings

Inspection Record 1

Equipment:	Hydraulic Press HP-200
Finding:	Severe corrosion was observed on the lower support frame of the hydraulic press. The affected area requires immediate attention.
Pit depth:	Pit depth measured at 12 mm.
Risk Level:	High
Recommended Action:	Repair the affected surface and perform a detailed structural integrity assessment before continued operation.

Inspection Record 2

Equipment:	Storage Tank ST-104
Finding:	Localized pitting corrosion was observed on the external surface of the storage tank near the lower section.
Pit depth:	Pit depth measured at 8 mm.
Risk Level:	Medium
Recommended Action:	Monitor the affected area and perform surface treatment to prevent further corrosion.

Inspection Record 3

Equipment:	Pressure Vessel PV-301
Finding:	Surface degradation and corrosion were detected near the welded joint on the pressure vessel.
Pit depth:	Pit depth measured at 15 mm.
Risk Level:	High
Recommended Action:	Conduct a detailed non-destructive examination and evaluate the vessel before further operation.

Inspection Record 4

Equipment:	Cooling Pipeline CP-45
Finding:	Multiple small corrosion pits were identified along the external surface of the cooling pipeline.
Pit depth:	Pit depth measured at 5 mm.
Risk Level:	Low
Recommended Action:	Clean and coat the affected area and include the pipeline in the next preventive maintenance cycle.

Inspection Record 5

Equipment:	Air Compressor AC-110
Finding:	Corrosion was observed around the compressor mounting bracket. No visible deformation was detected during inspection.
Pit depth:	Pit depth measured at 6 mm.
Risk Level:	Medium

Recommended Action:	Remove surface corrosion and apply protective coating. Reinspect during the next maintenance interval.
---------------------	--

Inspection Record 6

Equipment:	Chemical Storage Vessel CSV-22
Finding:	Significant localized corrosion was identified near the bottom section of the chemical storage vessel.
Pit depth:	Pit depth measured at 18 mm.
Risk Level:	Critical
Recommended Action:	Immediately isolate the equipment and conduct a complete structural integrity assessment before returning to service.

Summary of Engineering Findings

Equipment	Finding	Pit Depth
Hydraulic Press HP-200	Severe corrosion was observed on the lower support frame of the hydraulic press. The affected area requires immediate inspection and repair.	12 mm
Storage Tank ST-104	Localized pitting corrosion was observed on the external surface of the storage tank near the lower section.	8 mm
Pressure Vessel PV-301	Surface degradation and corrosion were detected near the welded joint on the pressure vessel.	15 mm
Cooling Pipeline CP-45	Multiple small corrosion pits were identified along the external surface of the cooling pipeline.	5 mm
Air Compressor AC-110	Corrosion was observed around the compressor mounting bracket. No visible deformation was detected during the inspection.	3 mm
Chemical Storage Vessel CSV-22	Significant localized corrosion was identified near the bottom section of the chemical storage vessel.	18 mm

Overall Assessment

The inspection identified several instances of surface corrosion and pitting across different equipment. The maximum measured pit depth was 18 mm, recorded on the Chemical Storage Vessel CSV-22. Equipment with high or critical risk findings should undergo detailed engineering assessment before continued operation.

Overall Status: Requires Corrective Action

Maximum Pit Depth: 18 mm

Report Status: Open – Engineering Review Required